Universidad Autónoma de Chihuahua

Facultad de Ingeniería

Titulo

Consultas del Proyecto 2a. Evaluación

Materia

Base de Datos Avanzadas

Docente

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RESPUESTAS DE LAS 25 CONSULTAS, CODIGO Y EJECUCION

1. Obtener los autos de modelo 2024 en adelante con rendimiento de al menos 35 mpg. Mostrar solo la descripción, el año y el rendimiento, ordenados de mayor a menor rendimiento.

```
db.cars.aggregate(
[
    {
        "$match": {
            "details.year": {
                "$gte": 2024
            },
            "details.mpg": {
                "$gte": 35
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "año": "$details.year",
            "rendimiento": "$details.mpg"
        }
    },
    {
        "$sort": {
            "rendimiento": -1
        }
    }
]
)
```

Registros encontrados: 18

Descripcion	Año	Rendimiento
toyota camry le hybrid 2.5	2025	51
toyota corolla hybrid le 1.8	2024	50
lexus es 300h 2.5	2024	44
honda accord hybrid touring 2.0	2024	44
suzuki swift 1.2	2024	44
peugeot 2008 gt 1.2t	2024	40
toyota rav4 hybrid xle 2.5	2024	40
renault kardian 1.0t	2025	40
mitsubishi mirage g4 1.2	2024	39
mg 5 1.5	2024	38
ford maverick xlt hybrid 2.5	2024	37
honda civic ex 1.5t	2024	36
suzuki vitara 1.4t	2024	36
hyundai elantra sel 2.0	2024	36
toyota corolla le 2.0	2024	35
nissan versa sr 1.6	2024	35
hyundai creta 1.5	2024	35
volkswagen jetta se 1.5t	2024	35

2. Listar los autos de fabricantes europeos con motor de 4 cilindros y al menos 250 hp, mostrando descripción, país y potencia, ordenados por potencia descendente.

```
db.cars.aggregate(
[
    {
        "$match": {
            "details.cylinders": 4,
            "details.horsepower": {
                "$gte": 250
            }
        }
    },
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
```

```
    },
    {
        "$match": {
            "maker_info.continent": "europe"
        }
    },
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "pais": "$maker_info.country",
            "potencia": "$details.horsepower"
        }
    },
    {
        "$sort": {
            "potencia": -1
        }
    }
]
)
```

Registros encontrados: 13

Descripcion	Pais	Potencia
volkswagen golf r 2.0t	germany	315
volvo xc90 b6 2.0t	sweden	295
alfa romeo giulia 2.0t	netherlands	280
volkswagen atlas 2.0t	germany	269
dodge hornet gt 2.0t	netherlands	268
audi a4 45 tfsi 2.0t	germany	261
audi q5 45 tfsi 2.0t	germany	261
porsche macan t 2.0t	germany	261
bmw 330i 2.0t	germany	255
bmw x3 xdrive30i 2.0t	germany	255
mercedes-benz c 300 2.0t	germany	255
mercedes-benz e 350 2.0t	germany	255
mercedes-benz glc 300 2.0t	germany	255

3. Contar cuántos autos de fabricantes japoneses tienen motor turbo, es decir, cuya descripción termina con la cilindrada seguida de "t" (por ejemplo "1.5t").

```
db.cars.aggregate(
[
    {
        "$match": {
            "descr": {
                "$regex": "t$",
                "$options": "i"
            }
        }
    },
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$match": {
            "maker_info.country": "japan"
        }
    },
    {
        "$count": "total_autos_japoneses_turbo"
    }
]
)
```

Registros encontrados: 1

Total_autos_japonese:
11

4. Encontrar los autos que cumplan cualquiera de estas condiciones: (a) pesan menos de 3000 lb y aceleran de 0 a 60 mph en menos de 9 s, o (b) tienen 8 o más cilindros y rinden al menos 19 mpg.

```
db.cars.aggregate(
[
    {
        "$match": {
            "$or": [
                {
                    "details.weight": {
                        "$lt": 3000
                    },
                    "details.accel": {
                        "$lt": 9
                    }
                },
                {
                    "details.cylinders": {
                        "$gte": 8
                    },
                    "details.mpg": {
                        "$gte": 19
                    }
                }
            ]
        }
    },
    {
        "$project": {
            "_id": 0,
            "descr": 1,
            "details.weight": 1,
            "details.accel": 1,
            "details.cylinders": 1,
            "details.mpg": 1
        }
    }
]
)
```

```
[
    {
        "descr": "chevrolet corvette stingray 6.2",
        "details": {
            "mpg": 19,
            "cylinders": 8,
            "weight": 3650,
            "accel": 2.9
        }
    },
    {
        "descr": "chevrolet camaro ss 6.2",
        "details": {
            "mpg": 19,
            "cylinders": 8,
            "weight": 3685,
            "accel": 4
        }
    },
    {
        "descr": "peugeot 2008 gt 1.2t",
        "details": {
            "mpg": 40,
            "cylinders": 3,
            "weight": 2800,
            "accel": 8.7
        }
    },
    {
        "descr": "toyota gr86 2.4",
        "details": {
            "mpg": 22,
            "cylinders": 4,
            "weight": 2811,
            "accel": 6.1
        }
    },
```

```
    {
        "descr": "mazda mx-5 miata 2.0",
        "details": {
            "mpg": 30,
            "cylinders": 4,
            "weight": 2350,
            "accel": 5.7
        }
    },
    {
        "descr": "hyundai elantra sel 2.0",
        "details": {
            "mpg": 36,
            "cylinders": 4,
            "weight": 2900,
            "accel": 8.5
        }
    },
    {
        "descr": "kia forte lxs 2.0",
        "details": {
            "mpg": 34,
            "cylinders": 4,
            "weight": 2900,
            "accel": 8.6
        }
    },
    {
        "descr": "mini cooper s 2.0t",
        "details": {
            "mpg": 31,
            "cylinders": 4,
            "weight": 2900,
            "accel": 6.4
        }
    }
]
```

5. Obtener los fabricantes que comercializan más de una marca o línea de modelo, mostrando nombre, país y modelos.

```
db.makers.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "_id",
            "foreignField": "maker_id",
            "as": "models_list"
        }
    },
    {
        "$match": {
            "$expr": {
                "$gt": [
                    {
                        "$size": "$models_list"
                    },
                    1
                ]
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "nombre": "$maker",
            "pais": "$country",
            "modelos": "$models_list.model"
        }
    }
]
)
```

```
[
    {
        "nombre": "gm",
        "pais": "usa",
        "modelos": [
            "chevrolet",
            "gmc",
            "cadillac",
            "buick"
        ]
    },
    {
        "nombre": "ford",
        "pais": "usa",
        "modelos": [
            "ford",
            "lincoln"
        ]
    },
    {
        "nombre": "stellantis",
        "pais": "netherlands",
        "modelos": [
            "jeep",
            "dodge",
            "ram",
            "chrysler",
            "fiat",
            "peugeot",
            "alfa romeo"
        ]
    },
```

```
    {
        "nombre": "toyota",
        "pais": "japan",
        "modelos": [
            "toyota",
            "lexus"
        ]
    },
    {
        "nombre": "honda",
        "pais": "japan",
        "modelos": [
            "honda",
            "acura"
        ]
    },
    {
        "nombre": "nissan",
        "pais": "japan",
        "modelos": [
            "nissan",
            "infiniti"
        ]
    },
    {
        "nombre": "hyundai",
        "pais": "korea",
        "modelos": [
            "hyundai",
            "kia",
            "genesis"
        ]
    },
```

```
    {
        "nombre": "volkswagen",
        "pais": "germany",
        "modelos": [
            "volkswagen",
            "audi",
            "porsche",
            "lamborghini",
            "bentley"
        ]
    },
    {
        "nombre": "bmw",
        "pais": "germany",
        "modelos": [
            "bmw",
            "mini"
        ]
    },
    {
        "nombre": "renault",
        "pais": "france",
        "modelos": [
            "renault",
            "dacia"
        ]
    },
    {
        "nombre": "jlr",
        "pais": "uk",
        "modelos": [
            "land rover",
            "jaguar"
        ]
    }
]
```

6. Encontrar los autos cuya marca (model.name) es distinta del nombre de su grupo fabricante (maker.name). Mostrar descripción, marca y fabricante.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$match": {
            "$expr": {
                "$ne": [
                    "$model_info.model",
                    "$maker_info.maker"
                ]
            }
        }
    },
```

```
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "marca": "$model_info.model",
            "fabricante": "$maker_info.maker"
        }
    }
]
)
```

Registros encontrados: 60

Descripcion	Marca	Fabricante
chevrolet silverado 1500 lt 5.3	chevrolet	gm
chevrolet tahoe lt 5.3	chevrolet	gm
chevrolet equinox lt 1.5t	chevrolet	gm
chevrolet trax 1rs 1.2t	chevrolet	gm
chevrolet malibu lt 1.5t	chevrolet	gm
chevrolet corvette stingray 6.2	chevrolet	gm
chevrolet camaro ss 6.2	chevrolet	gm
chevrolet colorado z71 2.7t	chevrolet	gm
gmc sierra 1500 denali 6.2	gmc	gm
gmc yukon denali 6.2	gmc	gm
gmc terrain sle 1.5t	gmc	gm
cadillac escalade 6.2	cadillac	gm
cadillac ct5-v blackwing 6.2	cadillac	gm
cadillac xt5 premium luxury 2.0t	cadillac	gm
buick envista 1.2t	buick	gm
buick enclave 2.5t	buick	gm
lincoln navigator 3.5	lincoln	ford
lincoln aviator 3.0	lincoln	ford
jeep wrangler rubicon 3.6	jeep	stellantis
jeep grand cherokee limited 3.6	jeep	stellantis
jeep compass latitude 2.0t	jeep	stellantis
jeep gladiator sport 3.6	jeep	stellantis
dodge durango r/t 5.7	dodge	stellantis
dodge charger scat pack 6.4	dodge	stellantis
dodge challenger srt hellcat 6.2	dodge	stellantis
dodge hornet gt 2.0t	dodge	stellantis
ram 1500 laramie 3.0 hurricane	ram	stellantis
chrysler pacifica touring l 3.6	chrysler	stellantis
fiat 500x sport 1.3t	fiat	stellantis
peugeot 2008 gt 1.2t	peugeot	stellantis
peugeot 3008 gt 1.6t	peugeot	stellantis
alfa romeo giulia 2.0t	alfa romeo	stellantis
alfa romeo stelvio quadrifoglio 2.9	alfa romeo	stellantis
lexus rx 350 2.4t	lexus	toyota
lexus es 300h 2.5	lexus	toyota
lexus lx 600 3.4	lexus	toyota
acura integra a-spec 1.5t	acura	honda
acura mdx 3.5	acura	honda
infiniti qx60 2.0t	infiniti	nissan
kia forte lxs 2.0	kia	hyundai

Descripcion	Marca	Fabricante
kia sportage lx 2.5	kia	hyundai
kia telluride sx 3.8	kia	hyundai
kia k5 gt 2.5t	kia	hyundai
kia sorento 2.5	kia	hyundai
genesis g70 3.3t	genesis	hyundai
genesis gv80 2.5t	genesis	hyundai
audi a4 45 tfsi 2.0t	audi	volkswagen
audi q5 45 tfsi 2.0t	audi	volkswagen
audi rs 6 avant 4.0	audi	volkswagen
porsche 911 carrera 3.0	porsche	volkswagen
porsche cayenne 3.0	porsche	volkswagen
porsche macan t 2.0t	porsche	volkswagen
mini cooper s 2.0t	mini	bmw
land rover defender 110 3.0	land rover	jlr
land rover range rover sport 3.0	land rover	jlr
jaguar f-pace p250 2.0t	jaguar	jlr
jaguar f-type r 5.0	jaguar	jlr
mg zs 1.5	mg	saic
mg 5 1.5	mg	saic
mg hs 1.5t	mg	saic

7. Calcular por país el número de autos, el rendimiento promedio y la potencia promedio (redondeados a 1 decimal), ordenado por número de autos.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$group": {
            "_id": "$maker_info.country",
            "num_autos": {
                "$sum": 1
            },
            "avg_mpg": {
                "$avg": "$details.mpg"
            },
            "avg_hp": {
                "$avg": "$details.horsepower"
            }
        }
    },
```

```
    {
        "$project": {
            "_id": 0,
            "pais": "$_id",
            "num_autos": 1,
            "rendimiento_promedio": {
                "$round": [
                    "$avg_mpg",
                    1
                ]
            },
            "potencia_promedio": {
                "$round": [
                    "$avg_hp",
                    1
                ]
            }
        }
    },
    {
        "$sort": {
            "num_autos": -1
        }
    }
]
)
```

Registros encontrados: 10

Num_autos	Pais	Rendimiento_promedi	Potencia_promedio
45	japan	29.6	218.2
28	usa	22.0	328.8
21	germany	25.0	302.7
15	netherlands	23.1	323.8
13	korea	26.9	235.5
5	china	32.0	130.4
4	uk	20.0	392.8
3	sweden	25.0	263.0
2	france	36.0	106.0
2	italy	15.5	715.5

8. Calcular para cada grupo fabricante el número de autos, el número de marcas distintas con autos registrados y la potencia promedio (redondeada a 1 decimal), ordenado por número de autos.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$group": {
            "_id": "$model_info.maker_id",
            "num_autos": {
                "$sum": 1
            },
            "marcas_distintas": {
                "$addToSet": "$model_info._id"
            },
            "avg_hp": {
                "$avg": "$details.horsepower"
            }
        }
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": "$maker_info.maker",
            "num_autos": 1,
            "num_marcas_distintas": {
                "$size": "$marcas_distintas"
            },
            "potencia_promedio": {
                "$round": [
                    "$avg_hp",
                    1
                ]
            }
        }
    },
    {
        "$sort": {
            "num_autos": -1
        }
    }
]
)
```

Registros encontrados: 20

Num_autos	Fabricante	Num_marcas_distintas	Potencia_promedio
16	gm	4	327.8
15	stellantis	7	323.8
14	toyota	2	255.3
13	hyundai	3	235.5
12	ford	2	330.1
12	volkswagen	3	285.5
8	honda	2	227.8
8	nissan	2	219.4
5	mazda	1	219.0
5	bmw	2	317.8
4	subaru	1	203.8
4	jlr	2	392.8
4	mercedes-benz	1	335.5
3	saic	1	128.3
3	volvo	1	263.0
3	suzuki	1	106.7
3	mitsubishi	1	146.7
2	renault	1	106.0
2	chery	1	133.5
2	ferrari	1	715.5

9. Contar cuántos modelos (marcas) hay por continente

```
db.models.aggregate(
[
    {
        "$lookup": {
            "from": "makers",
            "localField": "maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$group": {
            "_id": "$maker_info.continent",
            "total_modelos": {
                "$sum": 1
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "continente": "$_id",
            "total_modelos": 1
        }
    }
]
)
```

Registros encontrados: 3

Total_modelos	Continente
15	asia
6	america
21	europe

10. Clasificar los autos en rangos de rendimiento [0,20), [20,25), [25,30), [30,35), [35,40) y 40 o más, contando cuántos hay en cada rango y su peso promedio.

```
db.cars.aggregate(
[
    {
        "$bucket": {
            "groupBy": "$details.mpg",
            "boundaries": [
                0,
                20,
                25,
                30,
                35,
                40
            ],
            "default": "40 o más",
            "output": {
                "cantidad_autos": {
                    "$sum": 1
                },
                "peso_promedio": {
                    "$avg": "$details.weight"
                }
            }
        }
    }
]
)
```

Registros encontrados: 6

_id	Cantidad_autos	Peso_promedio
0	28	4748.928571428572
20	36	4328.083333333333
25	34	3686.764705882353
30	22	3102.2727272727275
35	10	2875.0
40 o más	8	3143.75

11. Obtener los 5 fabricantes con mayor rendimiento promedio, considerando solo aquellos con al menos 8 autos registrados.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$group": {
            "_id": "$model_info.maker_id",
            "avg_mpg": {
                "$avg": "$details.mpg"
            },
            "total_autos": {
                "$sum": 1
            }
        }
    },
    {
        "$match": {
            "total_autos": {
                "$gte": 8
            }
        }
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": "$maker_info.maker",
            "rendimiento_promedio": {
                "$round": [
                    "$avg_mpg",
                    2
                ]
            },
            "total_autos": 1
        }
    },
    {
        "$sort": {
            "rendimiento_promedio": -1
        }
    },
    {
        "$limit": 5
    }
]
)
```

Registros encontrados: 5

Total_autos	Fabricante	Rendimiento_promedi
14	toyota	30.36
8	honda	29.62
8	nissan	27.75
13	hyundai	26.92
12	volkswagen	24.92

12. Clasificar cada auto como "Alta" (mpg ≥ 30), "Media" (20 ≤ mpg < 30) o "Baja" (mpg < 20) y contar cuántos autos hay por continente y categoría.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$addFields": {
            "categoria": {
                "$switch": {
                    "branches": [
                        {
                            "case": {
                                "$gte": [
                                    "$details.mpg",
                                    30
                                ]
                            },
```

```
                        "then": "Alta"
                    },
                    {
                        "case": {
                            "$gte": [
                                "$details.mpg",
                                20
                            ]
                        },
                        "then": "Media"
                    }
                ],
                "default": "Baja"
            }
        }
    }
},
{
    "$group": {
        "_id": {
            "continente": "$maker_info.continent",
            "categoria": "$categoria"
        },
        "total_autos": {
            "$sum": 1
        }
    }
},
{
    "$project": {
        "_id": 0,
        "continente": "$_id.continente",
        "categoria": "$_id.categoria",
        "total_autos": 1
    }
}
]
)
```

Registros encontrados: 9

Total_autos	Continente	Categoria
26	europe	Media
10	america	Media
13	america	Baja
5	america	Alta
8	europe	Alta
27	asia	Alta
13	europe	Baja
2	asia	Baja
34	asia	Media

13. Desde la colección makers, obtener cuántos autos tiene registrados cada fabricante (incluidos los que tienen cero)

```
db.makers.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "_id",
            "foreignField": "maker_id",
            "as": "models_list"
        }
    },
    {
        "$lookup": {
            "from": "cars",
            "localField": "models_list._id",
            "foreignField": "model_id",
            "as": "cars_list"
        }
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": "$maker",
            "total_autos": {
                "$size": "$cars_list"
            }
        }
    }
]
)
```

Registros encontrados: 20

Fabricante	Total_autos
gm	16
ford	12
stellantis	15
toyota	14
honda	8
nissan	8
mazda	5
subaru	4
mitsubishi	3
suzuki	3
hyundai	13
volkswagen	12
bmw	5
mercedes-benz	4
renault	2
volvo	3
jlr	4
ferrari	2
saic	3
chery	2

14. Encontrar los modelos del catálogo que no tienen ningún auto registrado

```
db.models.aggregate(
[
    {
        "$lookup": {
            "from": "cars",
            "localField": "_id",
            "foreignField": "model_id",
            "as": "cars_list"
        }
    },
    {
        "$match": {
            "cars_list": {
                "$size": 0
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "modelo_sin_autos": "$model"
        }
    }
]
)
```

Registros encontrados: 3

Modelo_sin_autos
lamborghini
bentley
dacia

15. Para los 5 autos más potentes, obtener su descripción, potencia y el nombre completo (fullName) de su fabricante

```
db.cars.aggregate(
[
    {
        "$sort": {
            "details.horsepower": -1
        }
    },
    {
        "$limit": 5
    },
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "potencia": "$details.horsepower",
            "fabricante_fullName": "$maker_info.fullName"
        }
    }
]
)
```

Registros encontrados: 5

Descripcion	Potencia	Fabricante_fullname
ferrari 812 competizione	819	Ferrari N.V.
dodge challenger srt hell	717	Stellantis N.V.
cadillac ct5-v blackwing	668	General Motors
ferrari roma 3.9	612	Ferrari N.V.
audi rs 6 avant 4.0	591	Volkswagen Group

16. Obtener, para cada continente, los 3 autos con mayor rendimiento (descripción y mpg).

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$sort": {
            "details.mpg": -1
        }
    },
    {
        "$group": {
            "_id": "$maker_info.continent",
            "top_cars": {
                "$push": {
                    "descripcion": "$descr",
                    "mpg": "$details.mpg"
                }
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "continente": "$_id",
            "top_3_autos": {
                "$slice": [
                    "$top_cars",
                    3
                ]
            }
        }
    }
]
```

```
[
    {
        "continente": "europe",
        "top_3_autos": [
            {
                "descripcion": "peugeot 2008 gt 1.2t",
                "mpg": 40
            },
            {
                "descripcion": "renault kardian 1.0t",
                "mpg": 40
            },
            {
                "descripcion": "volkswagen jetta se 1.5t",
                "mpg": 35
            }
        ]
    },
    {
        "continente": "asia",
        "top_3_autos": [
            {
                "descripcion": "toyota camry le hybrid 2.5",
                "mpg": 51
            },
            {
                "descripcion": "toyota corolla hybrid le 1.8",
                "mpg": 50
            },
            {
                "descripcion": "lexus es 300h 2.5",
                "mpg": 44
            }
        ]
    },
    {
        "continente": "america",
        "top_3_autos": [
            {
                "descripcion": "ford maverick xlt hybrid 2.5",
                "mpg": 37
            },
            {
                "descripcion": "chevrolet malibu lt 1.5t",
                "mpg": 32
            },
            {
                "descripcion": "ford escape st-line 1.5t",
                "mpg": 30
            }
        ]
    }
]
```

17. Para cada fabricante con más de un modelo, determinar cuál de sus modelos tiene más autos registrados.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$group": {
            "_id": {
                "maker_id": "$model_info.maker_id",
                "model_name": "$model_info.model"
            },
            "count_autos": {
                "$sum": 1
            }
        }
    },
    {
        "$sort": {
            "_id.maker_id": 1,
            "count_autos": -1
        }
    },
    {
        "$group": {
            "_id": "$_id.maker_id",
            "total_modelos": {
                "$sum": 1
            },
            "modelos": {
                "$push": {
                    "model_name": "$_id.model_name",
                    "count_autos": "$count_autos"
                }
            }
        }
    },
```

```
    {
        "$match": {
            "total_modelos": {
                "$gt": 1
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "maker_id": "$_id",
            "modelo_con_mas_autos": {
                "$arrayElemAt": [
                    "$modelos",
                    0
                ]
            }
        }
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$project": {
            "fabricante": "$maker_info.maker",
            "modelo_con_mas_autos": 1
        }
    }
]
)
```

```
[
    {
        "modelo_con_mas_autos": {
            "model_name": "chevrolet",
            "count_autos": 8
        },
        "fabricante": "gm"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "dodge",
            "count_autos": 4
        },
        "fabricante": "stellantis"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "nissan",
            "count_autos": 7
        },
        "fabricante": "nissan"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "hyundai",
            "count_autos": 6
        },
        "fabricante": "hyundai"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "volkswagen",
            "count_autos": 6
        },
        "fabricante": "volkswagen"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "bmw",
            "count_autos": 4
        },
        "fabricante": "bmw"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "land rover",
            "count_autos": 2
        },
        "fabricante": "jlr"
    },
```

```
    {
        "modelo_con_mas_autos": {
            "model_name": "toyota",
            "count_autos": 11
        },
        "fabricante": "toyota"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "ford",
            "count_autos": 10
        },
        "fabricante": "ford"
    },
    {
        "modelo_con_mas_autos": {
            "model_name": "honda",
            "count_autos": 6
        },
        "fabricante": "honda"
    }
]
```

18. Calcular el porcentaje que representa cada continente sobre el total de autos, con 2 decimales.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$group": {
            "_id": "$maker_info.continent",
            "autos_continente": {
                "$sum": 1
            }
        }
    },
    {
        "$group": {
            "_id": null,
            "total_general": {
                "$sum": "$autos_continente"
            },
            "datos": {
                "$push": {
                    "continente": "$_id",
                    "autos": "$autos_continente"
                }
            }
        }
    },
    {
        "$unwind": "$datos"
    },
```

```
    {
        "$group": {
            "_id": "$maker_info.continent",
            "autos_continente": {
                "$sum": 1
            }
        }
    },
    {
        "$group": {
            "_id": null,
            "total_general": {
                "$sum": "$autos_continente"
            },
            "datos": {
                "$push": {
                    "continente": "$_id",
                    "autos": "$autos_continente"
                }
            }
        }
    },
    {
        "$unwind": "$datos"
    },
    {
        "$project": {
            "_id": 0,
            "continente": "$datos.continente",
            "porcentaje": {
                "$round": [
                    {
                        "$multiply": [
                            {
                                "$divide": [
                                    "$datos.autos",
                                    "$total_general"
                                ]
                            },
                            100
                        ]
                    },
                    2
                ]
            }
        }
    }
]
)
```

Registros encontrados: 3

Continente	Porcentaje
asia	45.65
america	20.29
europe	34.06

19. Listar los 5 autos que más superan el rendimiento promedio de los autos con su mismo número de cilindros, mostrando ese promedio y la diferencia.

```
db.cars.aggregate(
[
    {
        "$setWindowFields": {
            "partitionBy": "$details.cylinders",
            "output": {
                "avg_mpg": {
                    "$avg": "$details.mpg"
                }
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "cilindros": "$details.cylinders",
            "mpg_auto": "$details.mpg",
            "promedio_cilindros": {
                "$round": [
                    "$avg_mpg",
                    2
                ]
            },
            "diferencia": {
                "$round": [
                    {
                        "$subtract": [
                            "$details.mpg",
                            "$avg_mpg"
                        ]
                    },
                    2
                ]
            }
        }
    },
    {
        "$sort": {
            "diferencia": -1
        }
    },
    {
        "$limit": 5
    }
]
)
```

Registros encontrados: 5

Descripcion	Cilindros	Mpg_auto	Promedio_cilindros	Diferencia
toyota camry le hybrid 2.	4	51	28.84	22.16
toyota corolla hybrid le 1	4	50	28.84	21.16
honda accord hybrid tour	4	44	28.84	15.16
lexus es 300h 2.5	4	44	28.84	15.16
toyota rav4 hybrid xle 2.5	4	40	28.84	11.16

20. Calcular, por continente, el número de autos, cuántos tienen 8 o más cilindros y el porcentaje que representan (1 decimal).

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$group": {
            "_id": "$maker_info.continent",
            "total_autos": {
                "$sum": 1
            },
            "autos_v8_plus": {
                "$sum": {
                    "$cond": [
                        {
                            "$gte": [
                                "$details.cylinders",
                                8
                            ]
                        },
                        1,
                        0
                    ]
                }
            }
        }
    },
```

```
    {
        "$project": {
            "_id": 0,
            "continente": "$_id",
            "total_autos": 1,
            "autos_8_o_mas_cilindros": "$autos_v8_plus",
            "porcentaje": {
                "$round": [
                    {
                        "$multiply": [
                            {
                                "$divide": [
                                    "$autos_v8_plus",
                                    "$total_autos"
                                ]
                            },
                            100
                        ]
                    },
                    1
                ]
            }
        }
    }
]
)
```

Registros encontrados: 3

Total_autos	Continente	Autos_8_o_mas_cilind	Porcentaje
47	europe	8	17.0
28	america	9	32.1
63	asia	0	0.0

21. Calcular la potencia por cada 1000 lb de peso y mostrar los 10 autos con mayor valor, junto con su país.

```
db.cars.aggregate(
[
    {
        "$project": {
            "descr": 1,
            "model_id": 1,
            "potencia_por_1000lb": {
                "$multiply": [
                    {
                        "$divide": [
                            "$details.horsepower",
                            "$details.weight"
                        ]
                    },
                    1000
                ]
            }
        }
    },
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
```

```
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "pais": "$maker_info.country",
            "ratio_potencia_peso": {
                "$round": [
                    "$potencia_por_1000lb",
                    2
                ]
            }
        }
    },
    {
        "$sort": {
            "ratio_potencia_peso": -1
        }
    },
    {
        "$limit": 10
    }
]
)
```

Registros encontrados: 10

Descripcion	Pais	Ratio_potencia_peso
ferrari 812 competizione	italy	227.5
ferrari roma 3.9	italy	176.88
cadillac ct5-v blackwing (	usa	161.94
dodge challenger srt hell	netherlands	161.12
jaguar f-type r 5.0	uk	147.44
chevrolet corvette stingra	usa	135.62
bmw m3 competition 3.0	germany	128.97
audi rs 6 avant 4.0	germany	128.48
chevrolet camaro ss 6.2	usa	123.47
ford mustang gt 5.0	usa	123.08

22. Para cada fabricante, mostrar solo los modelos cuyo nombre es distinto al nombre del fabricante, descartando a los fabricantes que no tengan ninguno.

```
db.makers.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "_id",
            "foreignField": "maker_id",
            "as": "models_list"
        }
    },
    {
        "$project": {
            "fabricante": "$maker",
            "modelos_filtrados": {
                "$filter": {
                    "input": "$models_list",
                    "as": "m",
                    "cond": {
                        "$ne": [
                            "$$m.model",
                            "$maker"
                        ]
                    }
                }
            }
        }
    },
    {
        "$match": {
            "modelos_filtrados.0": {
                "$exists": true
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": 1,
            "modelos_distintos": "$modelos_filtrados.model"
        }
    }
]
)
```

```
[
    {
        "fabricante": "gm",
        "modelos_distintos": [
            "chevrolet",
            "gmc",
            "cadillac",
            "buick"
        ]
    },
    {
        "fabricante": "ford",
        "modelos_distintos": [
            "lincoln"
        ]
    },
    {
        "fabricante": "stellantis",
        "modelos_distintos": [
            "jeep",
            "dodge",
            "ram",
            "chrysler",
            "fiat",
            "peugeot",
            "alfa romeo"
        ]
    },
    {
        "fabricante": "toyota",
        "modelos_distintos": [
            "lexus"
        ]
    },
    {
        "fabricante": "honda",
        "modelos_distintos": [
            "acura"
        ]
    },
    {
        "fabricante": "nissan",
        "modelos_distintos": [
            "infiniti"
        ]
    },
```

```
    {
        "fabricante": "hyundai",
        "modelos_distintos": [
            "kia",
            "genesis"
        ]
    },
    {
        "fabricante": "volkswagen",
        "modelos_distintos": [
            "audi",
            "porsche",
            "lamborghini",
            "bentley"
        ]
    },
    {
        "fabricante": "bmw",
        "modelos_distintos": [
            "mini"
        ]
    },
    {
        "fabricante": "renault",
        "modelos_distintos": [
            "dacia"
        ]
    },
    {
        "fabricante": "jlr",
        "modelos_distintos": [
            "land rover",
            "jaguar"
        ]
    },
    {
        "fabricante": "saic",
        "modelos_distintos": [
            "mg"
        ]
    }
]
```


23. En una sola consulta, obtener: (a) el número de autos por continente, (b) el número de autos por cilindros y (c) las estadísticas globales de mpg (mínimo, máximo y promedio).

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$facet": {
            "por_continente": [
                {
                    "$group": {
                        "_id": "$maker_info.continent",
                        "total": {
                            "$sum": 1
                        }
                    }
                }
            ],
            "por_cilindros": [
                {
                    "$group": {
                        "_id": "$details.cylinders",
                        "total": {
                            "$sum": 1
                        }
                    }
                }
            ],
```

```
            "estadisticas_mpg": [
                {
                    "$group": {
                        "_id": null,
                        "min_mpg": {
                            "$min": "$details.mpg"
                        },
                        "max_mpg": {
                            "$max": "$details.mpg"
                        },
                        "avg_mpg": {
                            "$avg": "$details.mpg"
                        }
                    }
                }
            ]
        }
    }
]
)
```

```
[
    {
        "por_continente": [
            {
                "_id": "asia",
                "total": 63
            },
            {
                "_id": "america",
                "total": 28
            },
            {
                "_id": "europe",
                "total": 47
            }
        ],
        "por_cilindros": [
            {
                "_id": 3,
                "total": 9
            },
            {
                "_id": 8,
                "total": 16
            },
            {
                "_id": 4,
                "total": 83
            },
            {
                "_id": 12,
                "total": 1
            },
            {
                "_id": 6,
                "total": 29
            }
        ],
        "estadisticas_mpg": [
            {
                "_id": null,
                "min_mpg": 14,
                "max_mpg": 51,
                "avg_mpg": 25.992753623188406
            }
        ]
    }
]
```


24. Asignar a cada auto alemán su posición por potencia dentro de su país y mostrar los 5 primeros.

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$match": {
            "maker_info.country": "germany"
        }
    },
    {
        "$sort": {
            "details.horsepower": -1
        }
    },
    {
        "$limit": 5
    },
    {
        "$project": {
            "_id": 0,
            "descripcion": "$descr",
            "potencia": "$details.horsepower",
            "pais": "$maker_info.country"
        }
    }
]
)
```

Registros encontrados: 5

Descripcion	Potencia	Pais
audi rs 6 avant 4.0	591	germany
mercedes-benz amg g 6:	577	germany
bmw m3 competition 3.0	503	germany
porsche 911 carrera 3.0	379	germany
bmw x5 xdrive40i 3.0	375	germany

25. Obtener para cada fabricante europeo su auto más eficiente

```
db.cars.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "model_id",
            "foreignField": "_id",
            "as": "model_info"
        }
    },
    {
        "$unwind": "$model_info"
    },
    {
        "$lookup": {
            "from": "makers",
            "localField": "model_info.maker_id",
            "foreignField": "_id",
            "as": "maker_info"
        }
    },
    {
        "$unwind": "$maker_info"
    },
    {
        "$match": {
            "maker_info.continent": "europe"
        }
    },
    {
        "$sort": {
            "details.mpg": -1
        }
    },
    {
        "$group": {
            "_id": "$maker_info._id",
            "fabricante": {
                "$first": "$maker_info.maker"
            },
            "auto_mas_eficiente": {
                "$first": "$descr"
            },
            "mayor_mpg": {
                "$first": "$details.mpg"
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": 1,
            "auto_mas_eficiente": 1,
            "mayor_mpg": 1
        }
    }
]
)
```